Question ID: cebfb2c6

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Medium

Question

Dated Ages of Lunar Samples from Select Missions

Mission name	Year	Landing site	Approximate age of lunar samples (billions of years)
Apollo 11	1969	Mare Tranquillitatis	3.6
Apollo 15	1971	Mare Imbrium	3.3
Apollo 17	1972	Mare Serenitatis	3.8
Chang'e 5	2020	Oceanus Procellarum	2.0

The Apollo program missions were spaceflights to the moon led by the United States during the 1960s and 1970s during which astronauts collected some samples of the moon's surface. More recently, China launched the Chang'e 5 mission, which returned additional lunar surface samples. Researchers have analyzed and dated each of the samples, concluding that the lunar samples collected during the Chang'e 5 mission are significant because ______

Which choice most effectively uses data from the table to complete the claim?

Answer

A. they are much younger than the samples brought back from any of the Apollo missions.

B. they were collected from the same landing site as the Apollo 11 mission.

C. they are closest in age to the samples brought back by the Apollo 17 mission.

D. they helped confirm the predicted ages of the lunar samples from the Apollo missions.

Correct Answer: A

Rationale

Choice A is the best answer because it effectively uses data from the table to complete the claim about the significance of the Chang'e 5 lunar samples. The table shows the approximate ages of lunar samples from four different missions: three Apollo missions and the Chang'e 5 mission. The Chang'e 5 samples are said to be approximately 2 billion years old, while the Apollo samples are each said to be more than 3 billion years old. In other words, based on the data in the table, the Chang'e 5 samples are much younger than those from the Apollo missions.

Choice B is incorrect because the table shows that the Chang'e 5 samples were taken from a landing site at Oceanus Procellarum, which none of the Apollo missions are shown to have visited. Choice C is incorrect because the table shows the Apollo 17 samples as approximately 3.8 billion years old, the Apollo 15 samples as approximately 3.3 billion years old, and the Chang'e 5 samples as approximately 2 billion years old, and therefore, the Chang'e samples are closer in age to the Apollo 15 samples than they are to the Apollo 17 samples. Choice D is incorrect because nothing in the text or table suggests that the Chang'e 5 samples were used to confirm the ages of the Apollo samples.